

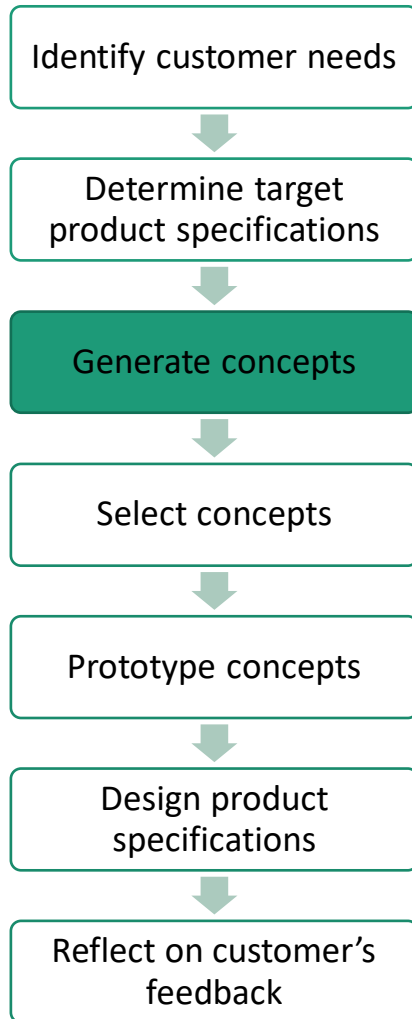
Concept Generation

Lecture 4

Outline

- Strategies for Creative Thinking:
 - Storyboarding and User Journey
 - Brainstorming
 - Analogy
 - SCAMPER
 - Morphology
 - Checklist questions
 - Biomimicry
 - Mind mapping
- Classroom Activity:
 - Getting started with Figma
 - Frames, layers, shapes and components in Figma

Generate Concepts

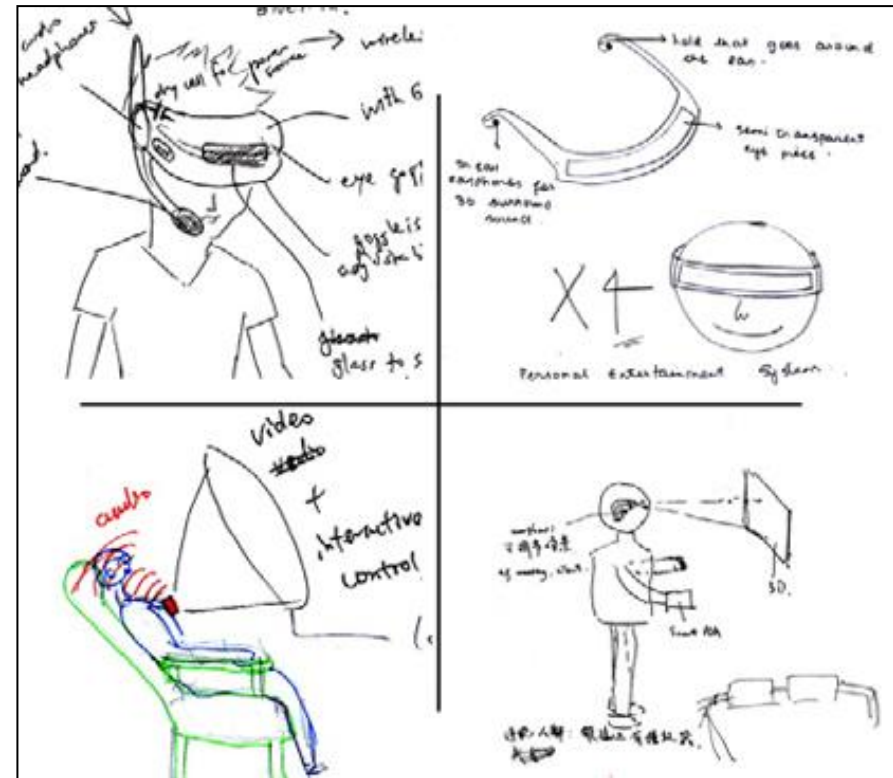
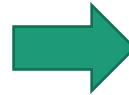


- Product concept is a technical description of how the product will satisfy customer's needs
- It should explain the technology to be applied, working principle and the approximate form of the product
- Often expressed with a sketch or 3D model accompanied with a textual description
- Various concepts are generated using common ideation technique
- Concepts are evaluated for their validity and continuously improved
- At early stages of concept generation phase, quantity is more important than quality.
- At later stages of concept generation phase, quality will prevail.

Explore ideas with creative thinking

	Custom er importa nce	Performance measure			Size of range		Technical details		
		Meet standa rd	Harne ss weigh t	Web bing stren gth	No. of color s	No. of s izes	Paddin g thickn ess	No. of buckl es	No. of gear loop s
Easy to put on	2					○		●	
Comfortable when hanging	5					○	●	○	
Fits over different clothes	1					○	○	●	
Accessible gear loops	3								●
Does not restrict movement	5		○			○	●	○	
Light weight	3		●	○			○	△	△
Safe	5	●	○	●					
Attractive	2		△		●		△	△	
Absolute importance		45	67	57	18	52	108	72	30

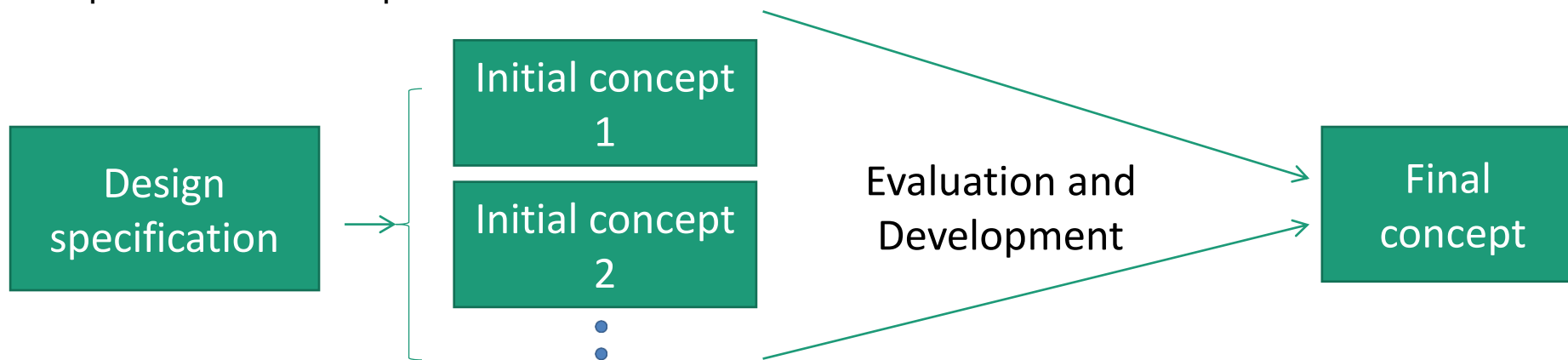
Design specifications



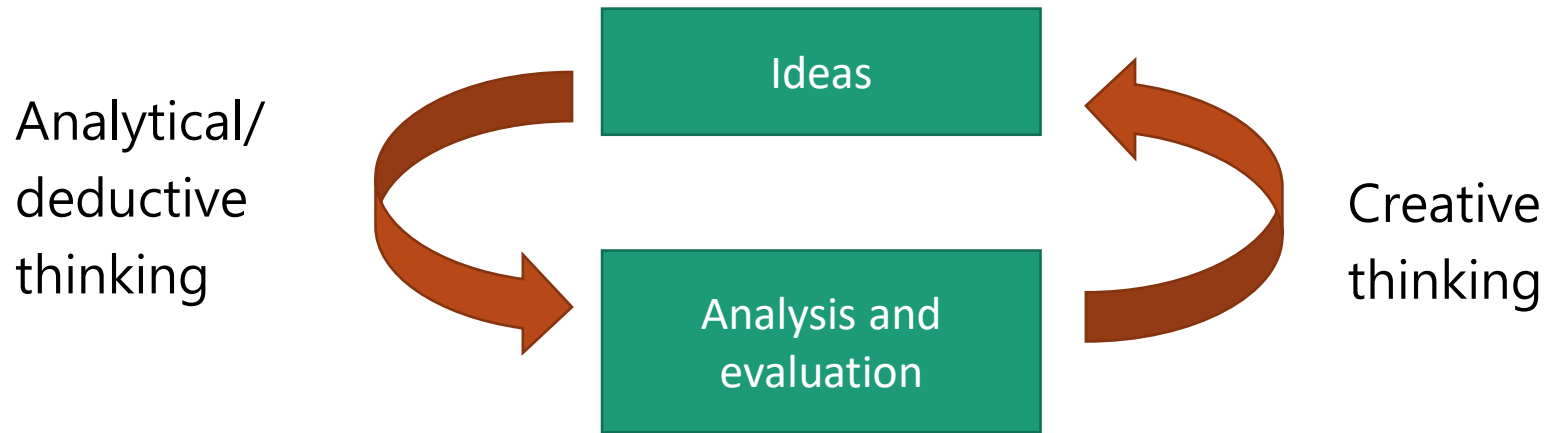
Conceptual designs

Concept Generation

- Initial phase of the product realization process
- Free-hand sketch to show vision or concept of the product, satisfying the design specification
- It is a high level overview but it also needs to be sufficiently detailed to allow, for example, estimates of cost, weight, overall dimensions, and power consumption
- Initial broad solutions (design options) are refined and optimized by iterations of evaluative and generative stages to gradually converge on a preferred conceptual solution



Design cycle



- Analytical/deductive thinking vs. Creative thinking
- Successful engineering design requires both.

Analytical vs. Creative

Analytical thinking

- Logic
- Convergent
- Leads to only one solution
- Considers only relevant information
- Implement scientific principle
- Pursue only realistic strategies
- Stop thinking when logic fails

Creative thinking

- Imagination, randomness, visualization, non-judgmental scrutiny
- Divergent
- Leads to many possible ideas
- Considers all information
- Stochastic, chaotic
- Continuously re-conceptualizes

Impediments to creativity

- Cultural impediments
 - Reluctant to ask
 - Prefer tradition
- Environmental impediments
 - Not proper physical environment
 - Colleagues who are not open to new ideas
- Emotional impediments
 - Reluctant to take risk
 - Do not want to develop ideas
 - Criticize ideas in early stages
 - Unmotivated
- Intellectual impediments
 - Limited education
 - Limited knowledge communication
 - Not exact information
- Technical impediments
 - No available computation tools
 - Not mature technologies
 - Underdeveloped manufacture process

Creativity killer

- Discouragement/No positive feedback
 - Difficult to continue working on new ideas when you have not received any positive feedback
- Self belief (I am not creative)
- Discomfort of 'being spatial' rather than 'thinking logically'
 - Trying to find the "right" answer
 - Being overly practical
- Risk aversion
 - Mistake? You can learn from it.

Strategies for creative thinking

- Storyboarding
- Analogy
- SCAMPER
- Morphology
- Checklist questions
- Biomimcry
- Mind mapping
- TRIZ (separate lecture)

Storyboarding and User Journey

1. Pick a Persona
2. Determine a Context for your Persona
3. Determine a Goal for your Persona
4. Determine the steps of achieving this goal using your application
5. Describe everything in User Journey Map
6. Sketch UI screens for each step in your Journey Map

Pick Persona as your User, Determine Context and Goals

User Persona

Problem statement:

Ejiro is a **busy artisan** who needs an **easy and less-techy way** to **order his meals** because he is **not digitally savvy**.



Ejiro

Age: 45
Education: SSCE
Hometown: Warri, Nigeria
Family: Divorced, Lives alone
Occupation: Plumber

“I like the idea of ordering my meals online, but the only issue is that I’m not tech-savvy”

Goals

- To live a stress-free lifestyle.
- To be productive and efficient at all times.

Frustrations

- “Food ordering apps should not be so complicated”
- “English is not my first language”.

Ejiro runs a plumbing company. As a new divorcee, he is struggling to get used to sourcing for his own meals, as his wife used to prepare his meals.

To make the process of getting his meals easier, he relies on restaurant mobile-ordering apps to order and receive his meals, especially when he is working. However, as someone who is not digitally savvy, he sometimes struggles with the meal ordering process. Language barrier is also a problem he faces as he is not fluent in English.



Pick Persona as your User, Determine Context and Goals

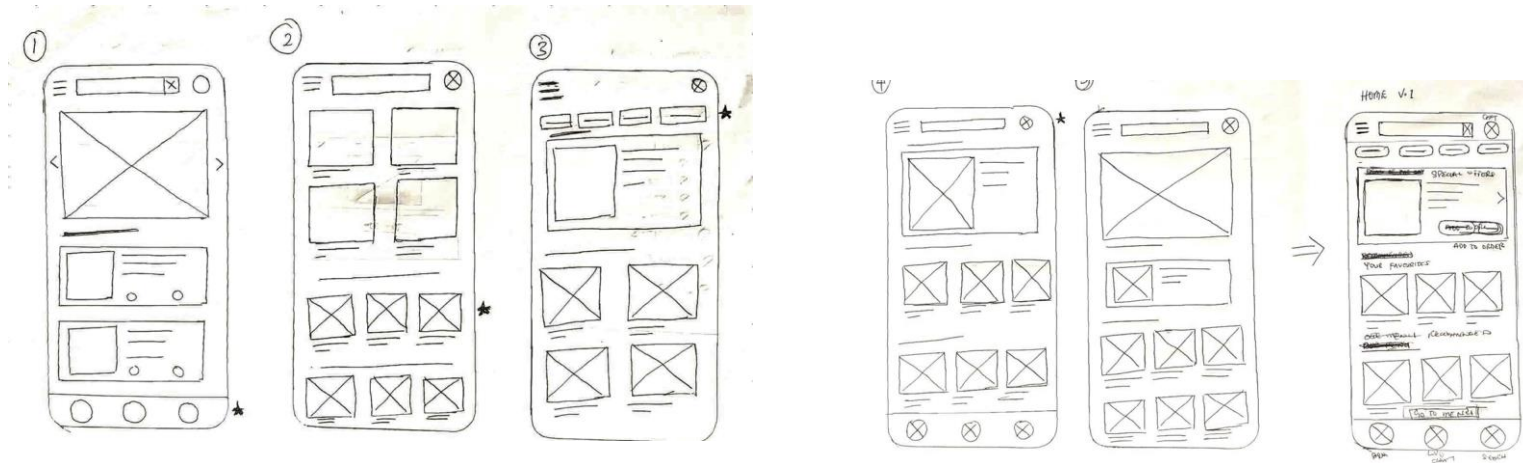
User Journey Map

Goal: Fast and easy food ordering via the Restaurant App

ACTION	Get App	Choose From Menu	Confirm Order	Checkout	Receive order
TASK LIST	Tasks A. Download App B. Create Account	Tasks A. Browse Menu B. Choose dishes C. Customise order	Tasks A. View cart B. Confirm total cost C. Confirm order	Tasks A. Enter delivery address B. Complete checkout C. Get order confirmation.	Tasks A. Make payment B. Check content of delivered items. C. Complete transaction.
FEELING ADJECTIVE	Excited about ordering lunch online. A bit stressed by the account creation process	Overwhelmed by the amount of dishes to choose from. Frustrated by the use of words & not enough images.	Excited about completing the ordering process. Stressed about struggling to read order details,	Worried about how long it would take to receive the order.	Happy about receiving order.
IMPROVEMENT OPPORTUNITIES	Make the onboarding process quick. Make the account creation process quick and easy.	Add a section with most popular dishes to the homepage. More images to accompany texts.	Make the cart easily accessible at all times. Make order details clear and legible.	Make it possible to track order. Add accurate estimated delivery time to confirmation email.	Add in-app feature that notifies the restaurant if there is an error in the order.

Paper wireframes

Initial wireframes by hand to encourage rapid iteration:



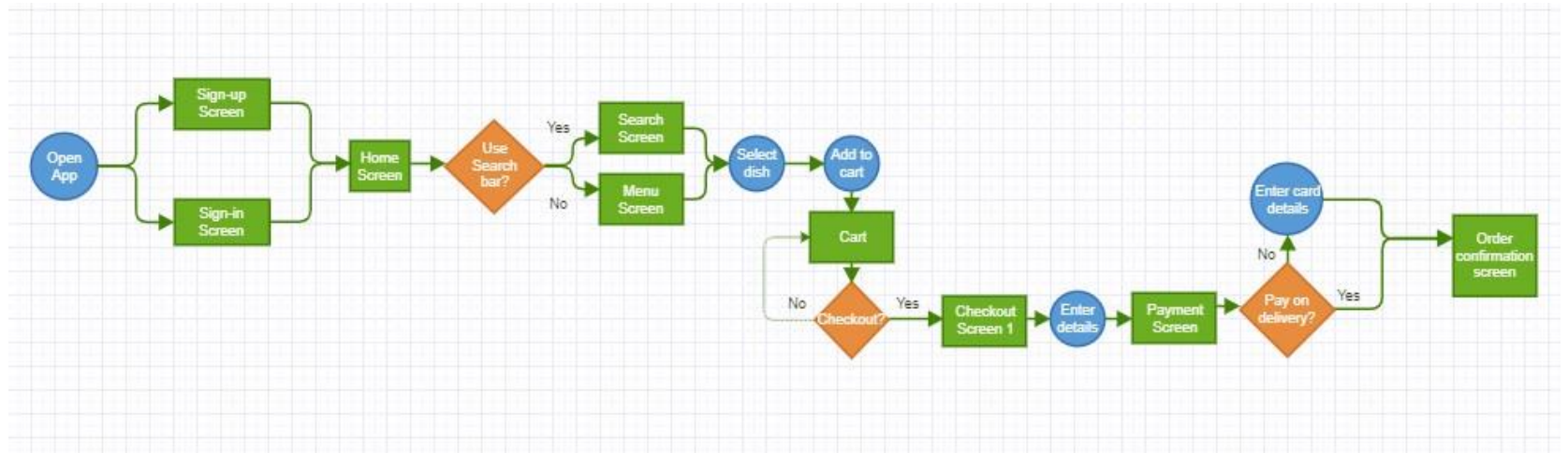
*Several iterations of each screen were drafted on paper to ensure that the elements that made it to the digital wireframes effectively addressed the user pain points.

User Flows

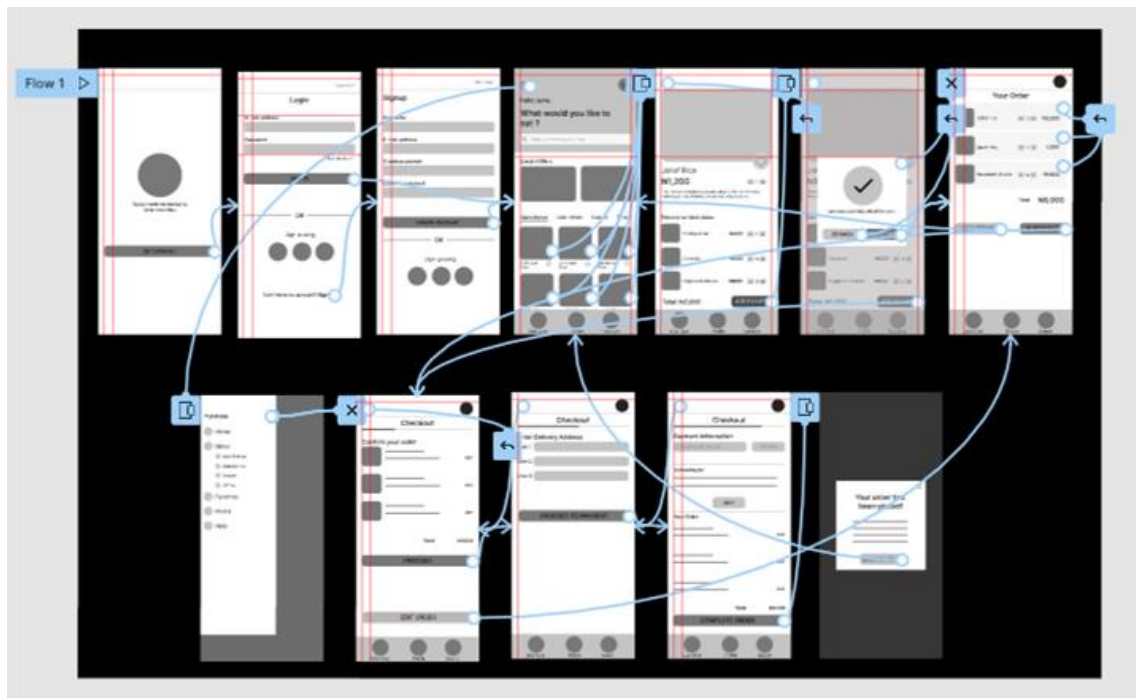
- It describes where users can navigate in your product.
- They allow you to understand how users interact with your app, the steps they take to complete a task or achieve a goal on your app.

	Marks the start/ end of the user flow
	Indicates steps users take
	Indicates a choice or decision point for the user
	The arrow indicates the flow of the symbol

User Flow for Each User Goal



Connecting the dots



SCAMPER

- Substitute.
- Combine.
- Adapt.
- Modify.
- Put to another use.
- Eliminate.
- Reverse

How to Use SCAMPER

- First, take an existing product or service. This could be one that you want to improve, one that you're currently having problems with, or one that you think could be a good starting point for future development.
- Then, ask questions about the product you identified, using the mnemonic to guide you. Brainstorm as many questions and answers as you can.
- Finally, look at the answers that you came up with. Do any stand out as viable solutions? Could you use any of them to create a new product, or develop an existing one? If any of your ideas seem viable, then you can explore them further.

Substitute

- What materials or resources can you substitute or swap to improve the product?
- What other product or process could you use?
- What rules could you substitute?
- Can you use this product somewhere else, or as a substitute for something else?
- What will happen if you change your feelings or attitude toward this product?

Substitute



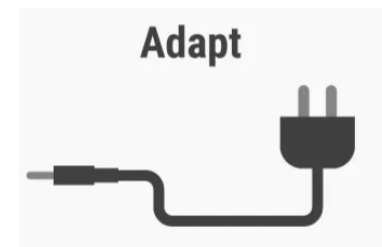
Combine

- What would happen if you combined this product with another, to create something new?
- What if you combined purposes or objectives?
- What could you combine to maximize the uses of this product?
- How could you combine talent and resources to create a new approach to this product?



Adapt

- How could you adapt or readjust this product to serve another purpose or use?
- What else is the product like?
- Who or what could you emulate to adapt this product?
- What else is like your product?
- What other context could you put your product into?
- What other products or ideas could you use for inspiration?



Modify (Magnify)

- How could you change the shape, look, or feel of your product?
- What could you add to modify this product?
- What could you emphasize or highlight to create more value?
- What element of this product could you strengthen to create something new?

Modify



Put to Another Use

- Can you use this product somewhere else, perhaps in another industry?
- Who else could use this product?
- How would this product behave differently in another setting?
- Could you recycle the waste from this product to make something new?

Put to another use



Eliminate

- How could you streamline or simplify this product?
- What features, parts, or rules could you eliminate?
- What could you understate or tone down?
- How could you make it smaller, faster, lighter, or more fun?
- What would happen if you took away part of this product? What would you have in its place?

Eliminate



Reverse

- What would happen if you reversed this process or sequenced things differently?
- What if you try to do the exact opposite of what you're trying to do now?
- What components could you substitute to change the order of this product?
- What roles could you reverse or swap?
- How could you reorganize this product?

Rearrange

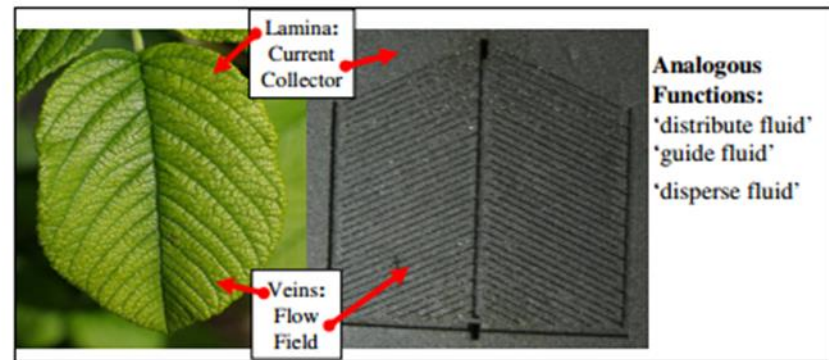
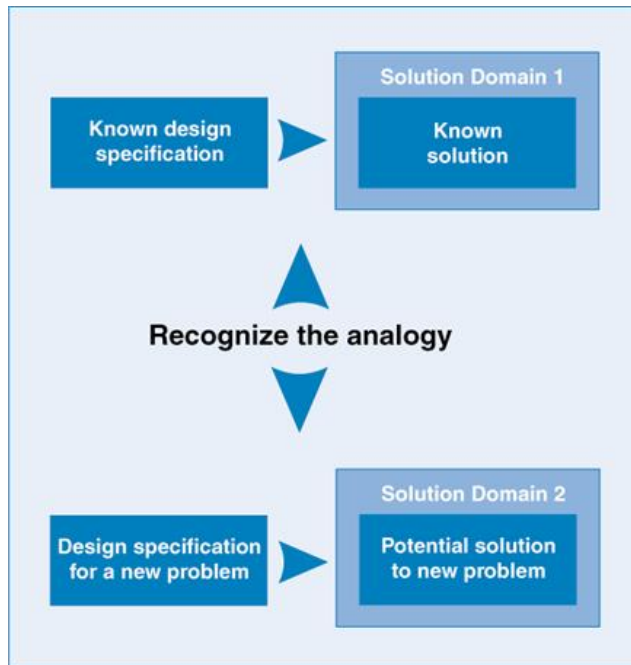


Exercise: Products/Services for SCAMPER

- Water bottle
- Scissors
- Pen or Pencil
- Eraser
- Bucket
- Barbershop
- Taxi
- Canteen
- Other daily items or services

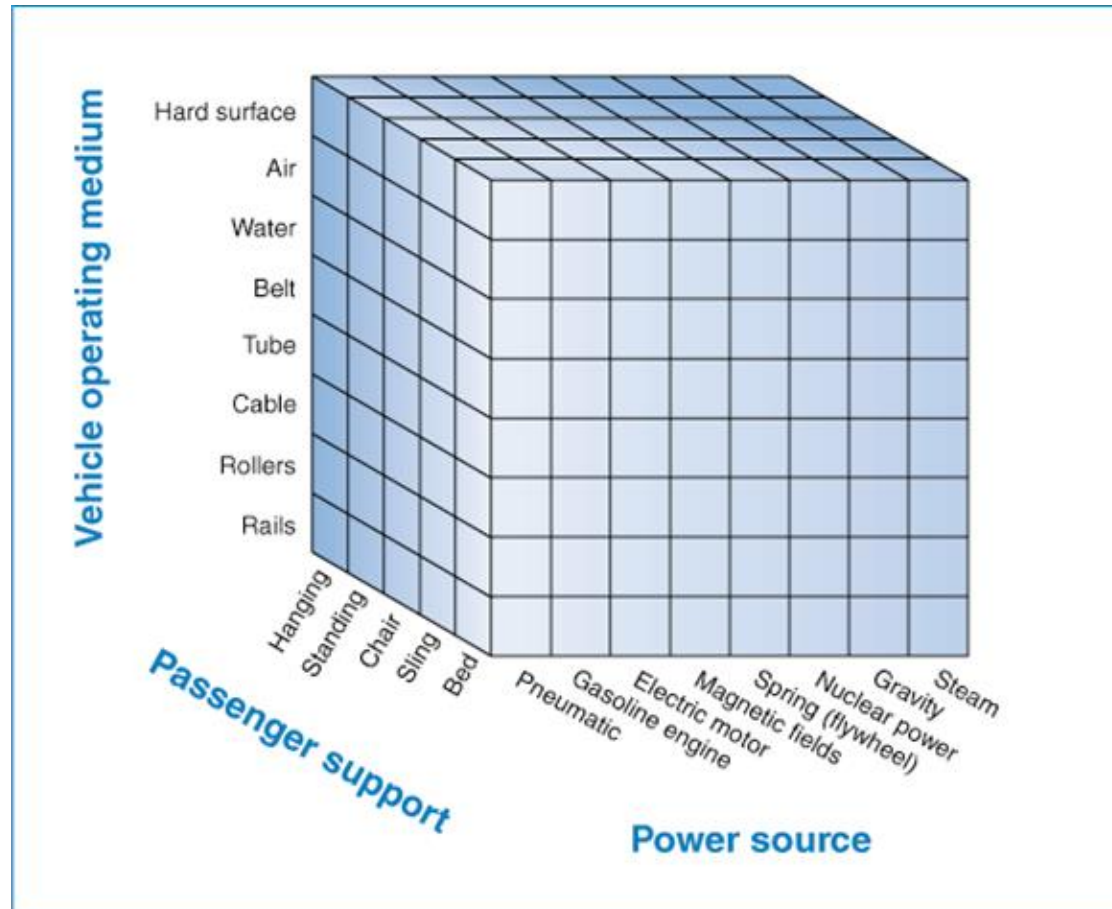
Strategy for creative thinking 2

- Analogy
 - Find similarity between two different domains



Strategy for creative thinking 3

- Morphology
 - Explore all possible combinations



Strategy for creative thinking 4

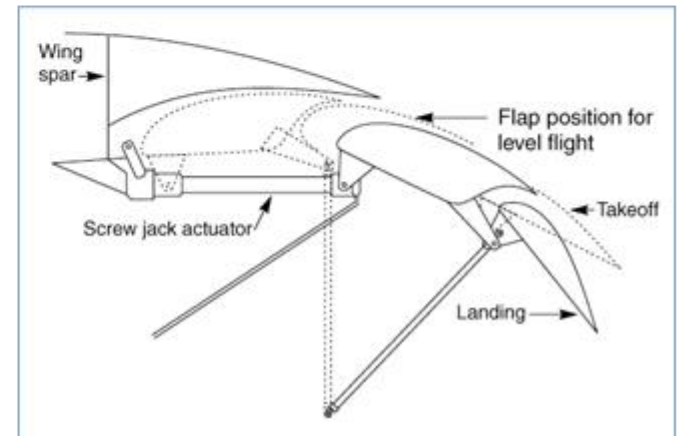
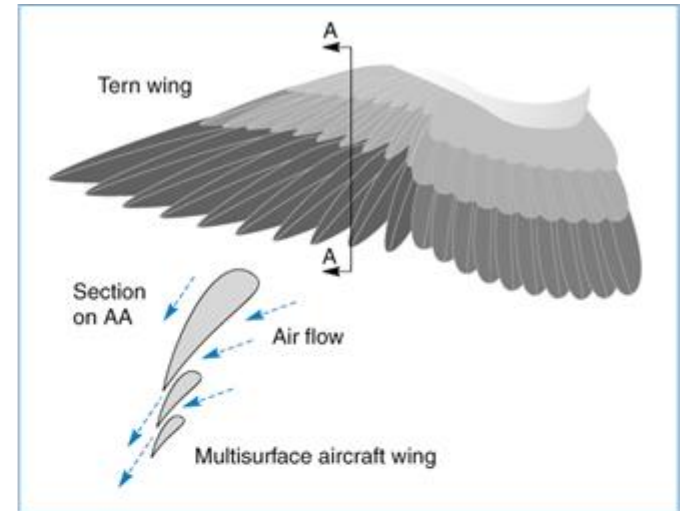
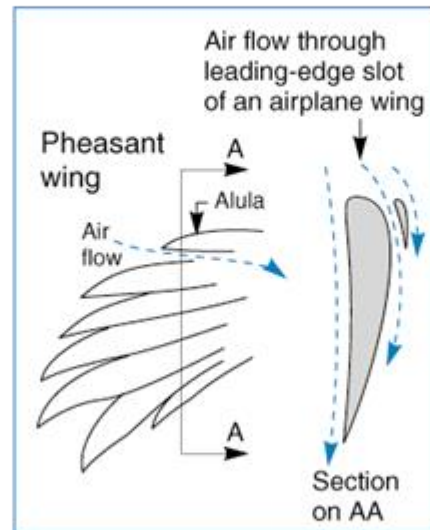
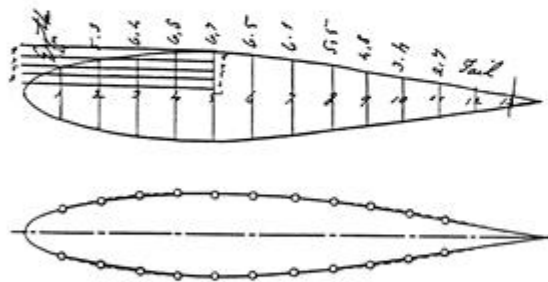
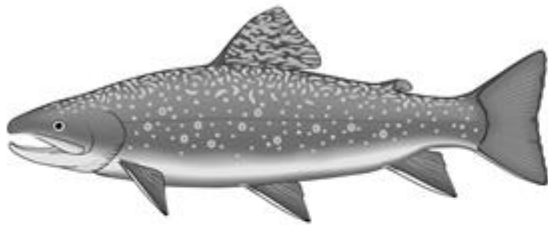
■ Checklist of questions

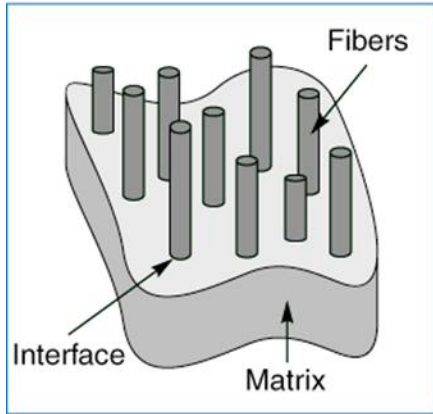
- Combination?
 - Combine units?
 - Combine objectives?
 - Combine VOCs?
 - Combine ideas?
- Invert?
 - Invert positive and negative poles?
 - What happen if we do it in inverted order?
 - Invert top and bottom?
 - Change the roles
- Realign?
 - Substitute parts?
 - In other shape?
 - Other alignment?
 - Different order?
 - Invert causal relation?
- Minimize?
 - What to remove?
 - Make it smaller?
 - Compress it?
 - Lower?
 - Shorter?
 - Lighter?
 - Thinner?
 - Split?
 - Reduce?
- Replace?
 - Replace with whom?
 - Replace with what?
 - Other ingredients?
 - Other process?
 - Other power source?
 - Other place?
 - Other approach?

A few examples

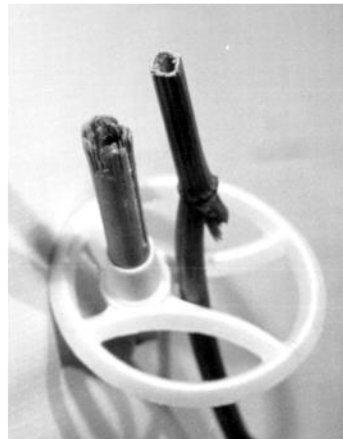
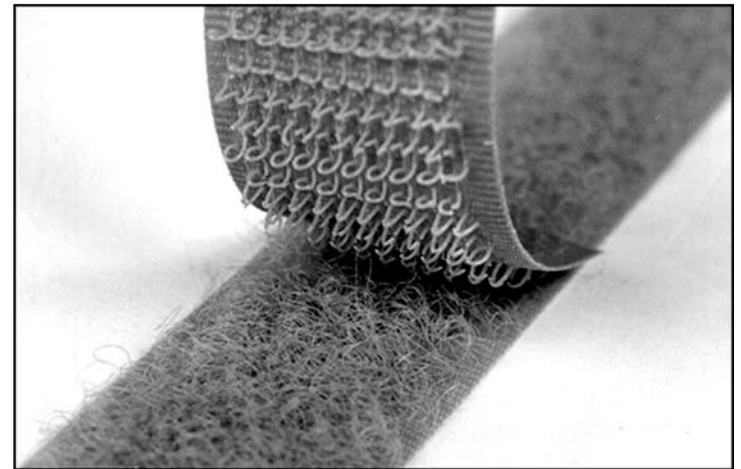
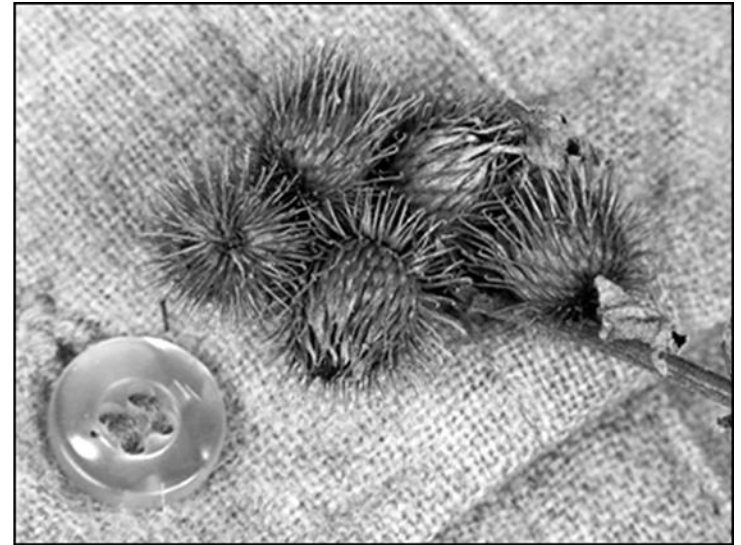
Strategy for creative thinking 5

- Biomimicry



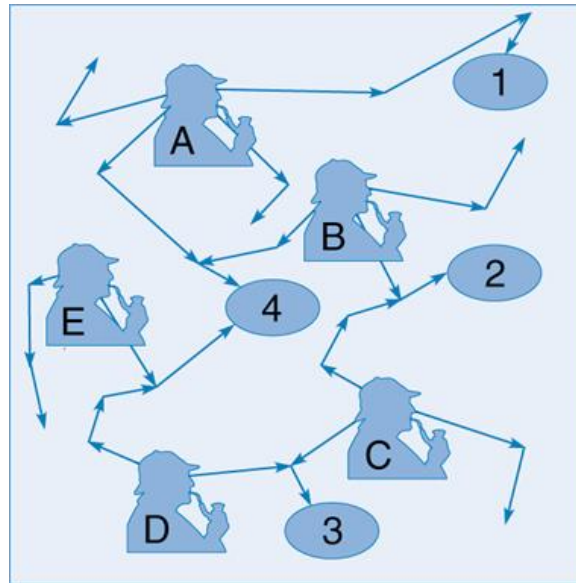


Coconut shell



Strategy for creative thinking 6

- Brainstorming

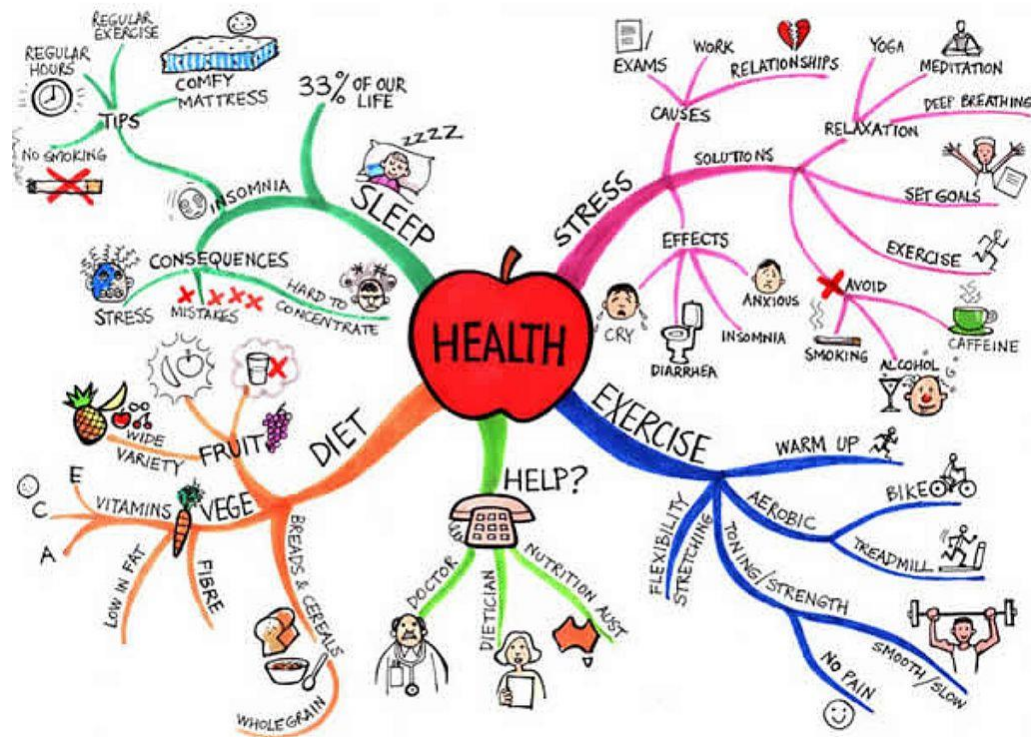


- Mind map (http://en.wikipedia.org/wiki/Mind_map)

Steps of Mind Mapping

1. Start in the center with an image of the topic, using at least 3 colors.
2. Use images, symbols, codes, and dimensions throughout your mind map.
3. Select key words and print using upper or lower case letters.
4. Each word/image is best alone and sitting on its own line.
5. The lines should be connected, starting from the central image. The lines become thinner as they radiate out from the center.
6. Make the lines the same length as the word/image they support.
7. Use multiple colors throughout the mind map, for visual stimulation and also for encoding or grouping.
8. Develop your own personal style of mind mapping.
9. Use emphasis and show associations in your mind map.
10. Keep the mind map clear by using radial hierarchy or outlines to embrace your branches.

Mind Map



Figma